Devansh Sharma

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Computer vision engineer building and evaluating perception models (object detection, segmentation, pose estimation, vision-language); currently building a camera + LiDAR perception stack for autonomous driving. NeurIPS 2026, ITSC 2024.

Education

M.Sc. Computer Vision, University of Central Florida — GPA 3.92 — 2025 – Dec 2026
M.Sc. Computer Science, Bowling Green State University — GPA 3.75 — 2021 – 2022
B.Eng. Computer Science, Manav Rachna International University — GPA 3.20 — 2015 – 2019

Research Publications

Redmond*, Tito Cruz*, **Sharma**, Azad, Mitra, Vyas, Rawat. *"MaRiO: Multi-agent Collaborative Reasoning via Shared Observations in MLLMs."* **NeurIPS 2026, Evaluations & Datasets Track** (*equal contribution)

- Built the evaluation harness for a large-scale, zero-shot benchmark of **23 multimodal LLMs**: the best (Gemini 2.5 Pro) reached 55.7% vs. 73.2% for humans, and most open-source models scored 30–42%.

Sharma, Rakib Sayem, Ibrahim. *"LLM Hallucination Detection via Low-rank Multilinear Decomposition."* **Under review, ICLR 2027**

- LLM hallucination detector that beat the supervised baseline HARP in **all 8 model–dataset settings** (+7.1 AUROC on average) and ACT-ViT in 6 of 8, with about 40× fewer projection parameters than PCA.

Sharma, Hade, Tian. *"Comparison of Deep Object Detectors on a New Vulnerable Pedestrian Dataset."* **IEEE ITSC 2024**

- Introduced and open-sourced BGVP, an object detection dataset of vulnerable pedestrians: 2,000 images with **5,932 manually annotated boxes** in 4 classes (children, elderly, people with disabilities, non-vulnerable).
- Fine-tuned COCO-pretrained YOLOv4, YOLOv5, YOLOX, Faster R-CNN, and EfficientDet via transfer learning (best mAP@0.5 of 0.80); every detector was weakest on elderly pedestrians and in dark scenes.

Experience

Graduate Researcher, *Center for Research in Computer Vision (CRCV), UCF* — Jan 2025 – Present

- Maintain a shared research codebase (Git, code review, unit tests, documentation) running reproducible, large-scale training and evaluation jobs on a **SLURM multi-GPU cluster**.

Data Operations Manager, *Radford University, VA* — Tableau dashboards for survey analytics — Jun 2024 – Aug 2024

Machine Learning Intern, *Fellowship.AI (Remote)* — Apr 2022 – Jun 2022

- In a cross-functional team, built an end-to-end real-time computer vision pipeline for human activity recognition on video streams (image classification with data augmentation), reaching **96.7% accuracy**.

Projects

Lookout: Real-Time AV Perception Stack (in progress) *ROS2, nuScenes, PyTorch, PointPillars, YOLO*

- Building a camera + LiDAR + radar perception stack for autonomous driving on ROS2: 3D object detection (PointPillars), 2D detection across 6 cameras, sensor fusion, and multi-object tracking, targeting real-time on a single consumer GPU.
- Built the nuScenes-to-ROS2 replay pipeline (rosbag2/MCAP: 6 cameras, LiDAR, 5 radars, full sensor calibration via TF) and the calibration and coordinate-transform layer for multi-sensor fusion.

Viewpoint Robustness of VLMs for Embodied Action *Qwen2.5-VL, InternVL3, LLaVA, AI2-THOR, A100 HPC*

- Large-scale robustness evaluation of 4 open vision-language models on ALFRED embodied AI tasks: 35 camera yaw/pitch shifts, **101,760 evaluations**, two-pass pipeline (A100 HPC inference, AI2-THOR simulator replay).
- Statistical testing (Wilcoxon) showed pitch far more disruptive than yaw and **model size uncorrelated with robustness** (InternVL3-8B ∼85% vs. Gemma 3-12B ∼15% baseline success).

SAM2 + Detection Backends for Open-Vocabulary Segmentation *SAM2, Grounding DINO, YOLOv11, CamVid*

- Coupled the SAM2 segmentation foundation model with Grounding DINO (zero-shot, open-vocabulary) and YOLOv11 (supervised) object detectors on CamVid street scenes.
- Supervised detection + SAM2 beat zero-shot text-prompted detection by **49%** Dice on urban driving scenes.

Skills

Languages & Core: Python, C++, SQL, PyTorch, TensorFlow, Hugging Face, Git, Linux, Algorithms & Data Structures
CV Tasks: Object Detection, Segmentation, Pose Estimation, Activity Recognition, Image Processing, Medical Imaging, Data Augmentation
CV Tools: OpenCV, YOLO, SAM2, Grounding DINO, HRNet, ViTs, DynUNet, SwinUNETR, MONAI, CLIP/SigLIP
Multimodal: Vision-Language Models (Qwen2.5-VL, InternVL3, LLaVA), LLMs, GPT/Gemini APIs, LoRA/PEFT, LangGraph
MLOps & Infra: Docker, FastAPI, MLflow, W&B, GitHub Actions, pytest, SLURM, AWS (Certified)